

Date: 20 Oct 2025 (Asia/Kolkata)

A) FOLDER STRUCTURE (KEY)

- app/main.py → FastAPI app (includes routers)
- app/settings.py → Paths (COMPANIES_ROOT, TEMPLATES_WORD_ROOT, TMP_ROOT...),

SMTP

- app/routers/master.py → Generate Master (MCA/PAN/INCORP → _master_*.json)
- app/routers/templates_word.py → Word template APIs (list/render/exists, auditors list)
- app/routers/mail.py → Auditor Consent & Directors Pack (preview/send)
- app/routers/mail_sla.py → SLA (preview/send; isolated from other mail)
- app/utils/render_docx.py → DocxTemplate rendering + context helpers
- processes/p01_mca_extract.py → MCA extractor (pdfminer.six → JSON/TXT)
- processes/p02_pan_extract.py → PAN extractor (OCR/regex → fields)
- templates/word/*.docx → ALL Word templates (see section B)
- frontend/index.html → UI (Master, Docs, Mailing, SLA panels)
- frontend/emails/*.html → Email templates (auditor_consent, directors_pack, sla)

B) WORD TEMPLATES (ALL)

1. bod_auditor_first_appointment.docx

- Purpose: Board Resolution for Appointment of First Auditor
- Variables: company.name, company.cin, company.pan, company.tan, company.email, date
directors[] (dynamic rows: name, din, role), auditor fields when applicable
- Output: companies/<Company>/bod_auditor_first_appointment_<TS>.docx

2. intimation_first_auditor_TEMPLATE.docx

- Purpose: Intimation letter to auditor
- Variables: company.*; auditor.name, membership_no, firm_name, address; date
- Output: companies/<Company>/intimation_first_auditor_TEMPLATE_<TS>.docx

3. proposal_first_auditor_TEMPLATE.docx

- Purpose: Proposal / engagement letter (scope & fees confirmation)
- Variables: company.*; auditor.*; date
- Output: companies/<Company>/proposal_first_auditor_TEMPLATE_<TS>.docx

4. auditor_consent_TEMPLATE.docx

- Purpose: Auditor consent (to be printed on auditor letterhead, signed)
- Variables: company.*; auditor.name, membership_no; date
- Output: companies/<Company>/auditor_consent_TEMPLATE_<TS>.docx

5. letter_head_TEMPLATE.docx (optional)

- Purpose: Blank page with company header (CIN/PAN/TAN/Email)
- Variables: company.*; date
- Output: companies/<Company>/letter_head_TEMPLATE_<TS>.docx

6. sla_agreement_TEMPLATE.docx

- Purpose: SLA agreement (company name/address merged)
- Variables: company.name, company.registered_address, date
- Output: companies/<Company>/sla_agreement_TEMPLATE_<TS>.docx

C) EMAIL TEMPLATES (ALL)

1) emails/email_auditor_consent.html

- Audience: Auditor
- Subject: "Consent to Act as Auditor – {{ company.name }}"
- Body: Banner + polite request to print/sign consent (Sec 139(6)); reply link
- Attachments: auditor_consent_TEMPLATE_<TS>.docx (or PDF)

2) emails/email_directors_pack.html

- Audience: Company Directors
- Subject: "Documents for Signature – {{ company.name }}"
- Body: Why required (BOD / Intimation / Proposal); signature instructions
- Attachments: BOD, Intimation, Proposal (latest generated for company)

3) emails/email_sla.html

- Audience: Company
- Subject: "SLA Agreement – {{ company.name }}"
- Body: Short note with banner; request to sign SLA
- Attachments: sla_agreement_TEMPLATE_<TS>.docx

Common Context for Emails:

- company: { name, cin, pan, tan, email, registered_address }
- auditor: { name, membership_no, firm_name, address, email? }
- contact: { email, phone, address } and reply_link
- Attachments resolved from /companies/<Company> latest files.

D) APIs (BACKEND)

MASTER

- POST /api/master/generate
Inputs (multipart): mca_file (req), pan_file (opt), incorp_file (opt), pan_number (opt), company_email (opt)
Output: { company_folder, master_path, mca_outputs, pan_outputs }
Notes: pulls email from MCA JSON if available; updates _master*.json

TEMPLATES WORD

- GET /api/templates/word → list available .docx templates
- GET /api/templates/word/auditors → rows from auditors_master_final.xlsx
- GET /api/templates/word/exists?company_name=... → which docs exist for company
- POST /api/templates/word/render → render a template_id for company (and auditor/date)

MAIL (CONSENT & DIRECTORS)

- POST /api/mail/preview/auditor → returns HTML preview (no send)
- POST /api/mail/preview/directors → returns HTML preview
- POST /api/mail/send/auditor → sends consent to auditor
- POST /api/mail/send/company → sends pack to company (BOD/Intimation/Proposal)

MAIL (SLA – separate router)

- POST /api/mail/sla/preview → returns HTML preview for SLA
- POST /api/mail/sla/send → sends SLA email to company with attachment

E) FRONTEND (frontend/index.html) – PANELS & CONTROLS

1) Generate Master

- Fields: MCA PDF (req), PAN PDF (opt), Incorporation Cert (opt), PAN Number (opt), Company Email (opt)
- Button: Generate Master → POST /api/master/generate
- Console logs: [ui] GEN ... [api] POST ... and status badges
- After success: refresh Companies dropdown (GET /api/companies), preselect created company

2) Documents Section

- Company search + dropdown (GET /api/companies); count hint; suggestions datalist
- Auditor dropdown loaded when any toggle ON (GET /api/templates/word/auditors)
- Toggles: BOD / Intimation / Proposal / Auditor Consent / Letter Head
- Generate Selected: POST /api/templates/word/render per selected template_id
- Displays “Open document” links with download URLs

3) Mailing Section

- Check readiness (calls /exists) and shows missing docs warning
- Preview buttons → open modal (iframe) with HTML (auditor/directors)
- Send buttons → POST to mail endpoints with logs + status

4) SLA Section (independent flow)

- Company search + select (same provider as Documents)
- Generate SLA → render docx
- Preview SLA mail (modal iframe) → /api/mail/sla/preview
- Send SLA → /api/mail/sla/send

F) DATA SOURCES & CONTEXT

- MCA JSON: companies/<Company>/outputs/mca_extract_<TS>.json (master_data.email, directors[])
- Master JSON: companies/<Company>/_master_<slug>_<TS>.json (central store)
- Auditors XLSX: auditors_master_final.xlsx (id, auditor_name, membership_no, email, firm, address, fy, etc.)

G) NAMING, PATHS & STATIC

- All outputs saved under companies/<Company>/...
- File names include <template_id>_<YYYYMMDD_HHMMSS>.docx
- Static serving for /companies/* and /assets/img/* (banner) enabled in main.py

H) TROUBLESHOOTING (BY DOC/EMAIL)

- BOD/Intimation/Proposal not generating → ensure template exists; auditor selected where required
- Consent letter “membership_no” missing → check XLSX mapping and selected auditor_id
- Email sent but wrong salutation → ensure auditor/company name/email resolved from XLSX or master
- SLA 404 on preview → confirm mail_sla router included and email_sla.html exists
- Master email empty → verify MCA JSON has master_data.email or provide company_email in form

I) QUICK HOW-TO (END-TO-END)

1. Generate Master

- Upload MCA (and optionally PAN/INCORP, PAN number, company email) → Generate
- Confirm success message and new company in the dropdown

2. Create Documents

- Select the company
- Toggle the documents to create (BOD / Intimation / Proposal / Consent / Letter Head)
- Pick an auditor (loads from XLSX) → Generate Selected
- Open and verify the generated files from the provided links

3. Send Emails

- Mailing panel: Check readiness
- Preview Auditor / Directors email (modal) → confirm content
- Send → watch console logs for SMTP status

4. SLA Flow

- SLA section: Select company → Generate SLA
- Preview SLA email → Send

J) NOTES ON FLEXIBILITY

- Directors: templates support 1..N directors (repeat rows)
 - Header: company name (single or two-line), line of CIN/PAN/TAN, line of Email – centered
 - Date: front-end date picker; backend defaults to “today” if not provided
 - Subject & HTML: editable email templates in frontend/emails/*.html
 - SMTP: configured in app/settings.py (or env) – swap credentials for production
- End of FULL summary –